

Technology #06

Neuromorphic Shunt Failure Predictor

Engineering Technology-Transfer Dossier

CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION

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Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Neuromorphic Shunt Failure Predictor

WHY DOES IT MATTER?

Shunt failure prediction is reactive (symptoms then surgery)

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 9 design inputs, 7 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in Machine Learning + Clinical Data. 6-step build plan with specific test articles and acceptance criteria. Target buyer: Medical device AI/digital health company; neuroshunt manufacturer with digital platform.

WHAT DOES THE BUYER GET?

ML predictor architecture (gradient boosting, 0 prior-art hits); Feature extraction specification (rolling 24h window); Clinical alert threshold ($P > 0.7$); Power dependency analysis (conditional on P-15-R1/P-16); This engineering dossier with data-pipeline-specific design chain

WHAT DOES THE BUYER HAVE TO BUILD?

Data partnership for 100+ patient-years training data; Production ML pipeline (training, validation, deployment); SaMD regulatory submission (510(k) or De Novo); Cybersecurity design (if connected device); Clinical validation evidence

WHAT IS THE NEXT DECISIVE EXPERIMENT?

P-13 ML predictor: Synthetic training dataset (clearly labeled)

WHAT WOULD MAKE US KILL IT?

No real failure dataset exists to train the model (CRITICAL BLOCKER per Article XXXVII)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 5 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 6-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: P-13 ML predictor: Synthetic training dataset (clearly labeled)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - P-13 ML predictor: Synthetic training dataset (clearly labeled) - 8 weeks

Basis: Build plan WP. Scope: Dataset volume + diversity. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - P-13 ML predictor: Real failure data collection protocol - 12-24 months (clinical timeline)

Basis: Build plan WP. Scope: Real failure event count. Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - P-13 ML predictor: Software V&V; per IEC 62304 - 16 weeks

Basis: Build plan WP. Scope: Software safety class + V&V; coverage. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: Sensor data available with sufficient quality and rate; Failure precursors exist in sensor data (UNKNOWN — needs evidence); ML model can learn from historical data; False positive rate tolerable clinically. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

5 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: P-13 ML predictor: Synthetic training dataset (clearly labeled). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: ML-based predictor using ICP and flow trends to forecast failure ≥ 12 h before symptoms

POTENTIAL VALUE DRIVER: Addresses: Shunt failure prediction is reactive (symptoms then surgery)

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: Medical device AI/digital health company; neuroshunt manufacturer with digital platform

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: Machine Learning + Clinical Data

Disciplines: Data engineering, ML engineering, Clinical informatics, Software engineering (SaMD)

Architecture: ML predictor using ICP + flow trends to forecast shunt failure 24h before symptoms

Subsystems:

SS-01: ICP sensor stream - Continuous $P_{ICP}(t)$ data

SS-02: Flow sensor stream - Continuous $F(t)$ data

SS-03: Feature extractor - Rolling window features (24h)

SS-04: ML predictor (gradient boosting) - $P(\text{fail within 24h})$ prediction

SS-05: Clinical alert interface - Alert if $P > \text{threshold}$

2. Mechanism Architecture

Physical Changes: NOT ESTABLISHED

Key Physics: NOT ESTABLISHED

3. Governing Engineering Model

Summary: Neuromorphic ML predictor for shunt failure prediction. Sensor streams (flow, pressure) processed by feature extractor + ML model to predict imminent failure.

Equations:

```
P(failure | features) = ML_model(features) [prediction]
features = extract(raw_sensor_streams) [feature extraction]
Alert if P(failure) > threshold [decision rule]
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Assumptions:

- Sensor data available with sufficient quality and rate
- Failure precursors exist in sensor data (UNKNOWN — needs evidence)
- ML model can learn from historical data
- False positive rate tolerable clinically

Boundary Conditions:

- Sensor sampling rate: ~1-10 Hz typical
- Prediction horizon: hours to days (target)
- False positive tolerance: UNKNOWN clinical threshold

Failure Regimes:

- Insufficient training data (rare failure events) -> model underfits
- Distribution shift (new patient population) -> model degrades
- Sensor data quality poor -> features unreliable
- False positive rate too high -> alert fatigue
- False negative (missed failure) -> patient harm

4. Design Inputs

ID	Input	Value	Origin
DI-001	Clinical need	Shunt failure prediction is reactive (symptoms then surgery). No predictive tool exists	DEFINED
DI-002	Functional requirement	ML-based predictor using ICP + flow trends to forecast failure 24h before symptoms	MODELLED
DI-003	AUC target	>= 0.80	MODELLED
DI-004	Lead time target	>= 12h before symptoms	MODELLED
DI-005	Training data	UNKNOWN — no real failure dataset exists yet (per Article XXXVII: synthetic data only, real loop WAITING_FOR_REAL)	UNKNOWN
DI-006	Power source	CONDITIONAL on P-15-R1 or P-16	MODELLED
DI-007	SaMD classification	UNKNOWN	UNKNOWN
DI-008	Software V&V (IEC 62304)	UNKNOWN — production software not yet developed	UNKNOWN
DI-009	Regulatory pathway (SaMD)	UNKNOWN — FDA SaMD framework likely applies	UNKNOWN

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	ML model architecture	MODELLED	
DO-002	Feature extraction specification	MODELLED	
DO-003	Clinical alert interface	CONCEPTUAL	
DO-004	Deployment architecture	ABSENT	Hardware platform; Data pipeline; Cybersecurity design; Cloud/on-device

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
Sensor sampling rate	MODELLED (~1-10 Hz)	Hz	R332 model	
Prediction horizon target	MODELLED (hours to days)	hours	R332 model	
Training data volume	UNKNOWN — needs real failures data (CRITICAL UNKNOWN)	samples	Unknown	
False positive rate target	UNKNOWN — clinical tolerance	%BD	Clinical requirement	
False negative rate target	UNKNOWN — clinical tolerance	%BD	Clinical requirement	
Model architecture	MODELLED (neuromorphic proposed)	pos	R332 mechanism	

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
Insufficient training data	Failure events rare; not enough examples at collection	Multiple samples at collection	Cross-validation on held-out data	UNKNOWN (clearly labeled as such in test set)
Distribution shift	New patient population differs from training data	Diversify training data + domain adaptation	Prospective validation on new population	UNKNOWN generalization performance
False positives (alert fatigue)	Model produces too many false positives	Threshold tuning + clinician feedback loop	Lead-back study	UNKNOWN acceptable false positive rate clinically
False negatives (missed failures)	Model misses true failures	Conservative threshold + redundancy	Clinical sensitivity study	UNKNOWN acceptable false negative rate clinically
Sensor data quality	Sensor drift, noise, or failure	Sensor quality monitoring + sensor fault injection test	Sensor fault injection test	UNKNOWN sensor data quality in vivo
Model drift over time	Model performance degrades as patient physiology drifts	Periodic retraining + drift monitoring	Long-term performance monitoring	UNKNOWN drift rate
Software defects	Bugs in feature extraction or logic	Extensive software V&V + code review	Software V&V testing	UNKNOWN residual defect rate

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
Insufficient training data	Rare failure events; no real dataset	Multi-center data collection + transfer learning	UNKNOWN generalization (early data ex)
Distribution shift	New population differs from training	Diverse training + domain adaptation + unknown	UNKNOWN generalization
False positives (alert fatigue)	Too many false alerts	Threshold tuning + clinician feedback	UNKNOWN acceptable rate clinically
False negatives (missed failures)	Model misses true failures	Conservative threshold + redundancy with NVA	UNKNOWN acceptable rate clinically
Sensor data quality	Sensor drift/noise/failure degrades input	Sensor monitoring + imputation + redundancy	UNKNOWN sensor quality in vivo
Model drift over time	Performance degrades as patient changes	Periodic retraining + drift monitoring	UNKNOWN drift rate
Software defects	Bugs in feature extraction or prediction	IEC 62304 V&V + code review + testing	UNKNOWN residual defect rate

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
V-001	Model AUC >= 0.80 on held-out test set	Cross-validation on synthetic dataset (per Article 58 Xv)	AUC >= 0.80 on synthetic data	NOT_TESTED (synthetic dataset not yet)
V-002	Lead time >= 12h before clinical symptoms	Retrospective evaluation on synthetic time series	Median lead time >= 12h	NOT_TESTED
V-003	False positive rate clinically tolerable	Bench evaluation with non-failure data	FP rate < threshold (UNKNOWN specific threshold — needs clinical input)	NOT_TESTED
V-004	Software V&V per IEC 62304	IEC 62304 software lifecycle	IEC 62304 compliance (Class B not)	NOT_TESTED

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VAL-001	Clinical failure prediction accuracy	Prospective clinical trial (IDE required)	Sensitivity/specificity clinically	NOT_PERFORMED (specific threshold for clinical)

11. Materials

NOT ESTABLISHED

12. Bill of Materials

NOT ESTABLISHED

13. Manufacturing

Status: NOT ESTABLISHED

Candidate Processes:

14. External Evidence

Source 1: Implantable Intracranial Pressure Sensor with Continuous

URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC10532732>

Excerpt: by Y Elsawaf · 2023 · Cited by 4 — an ambulatory intracranial pressure (ICP) monitoring device may assist in the detection of shunt failure without an invasive diagnostic workup.

Source 2: Evolving concepts in intracranial pressure monitoring

URL: <https://www.sciencedirect.com/science/article/pii/S1878747924001946>

Excerpt: by R Mathur · 2025 · Cited by 20 — Intraparenchymal ICP monitors are an alternative to EVDs and use implantable micro-transducers to measure local ICP at the tip of the probe placed into the ... Read more

Source 3: Early fault prediction and detection of hydrocephalus

URL: https://file.scirp.org/Html/7-9101515_28918.htm

Excerpt: Long-term monitoring of intracranial pressure (ICP) is limited by the lack of an implantable sensor with low drift as well as the power needed for the implanted ... Read more

Source 4: Australia OKs coin-sized sensor for hydrocephalus

URL: <https://www.mobihealthnews.com/news/anz/australia-oks-coin-sized-sensor-hydrocephalus-management>

Excerpt: Mar 15, 2026 — The implant enables real-time intracranial pressure monitoring , cutting hospital scans by over 60%, monitoring procedures by 90%, and reducing ... Read more

Source 5: Emerging Technology

URL: <https://www.hydroassoc.org/emerging-technology>

Excerpt: The sensor responds to ultrasound signals and can be read using standard or handheld ultrasound devices. It is designed for long-term or acute ICP monitoring in ... Read more

15. Transfer Boundary

YOU RECEIVE:

- + ML predictor architecture (gradient boosting, 0 prior-art hits)
- + Feature extraction specification (rolling 24h window)
- + Clinical alert threshold ($P > 0.7$)
- + Power dependency analysis (conditional on P-15-R1/P-16)
- + This engineering dossier with data-pipeline-specific design chain

YOU MUST DEVELOP:

- > Data partnership for 100+ patient-years training data
- > Production ML pipeline (training, validation, deployment)
- > SaMD regulatory submission (510(k) or De Novo)
- > Cybersecurity design (if connected device)
- > Clinical validation evidence

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- Real failure dataset for training (CRITICAL — no real data exists yet, per Article XXXVII synthetic vs real loop) — UNKNOWN
- Achievable prediction accuracy (sensitivity + specificity) — UNKNOWN
- Clinical tolerance for false positives and false negatives — UNKNOWN
- Sensor data quality in vivo — UNKNOWN
- Model drift rate over chronic deployment — UNKNOWN
- Regulatory pathway (Software as Medical Device, FDA SaMD guidance) — UNKNOWN specifics
- Generalization across patient populations — UNKNOWN

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	P-13 ML predictor: Synthetic training data (as defined)	Computer in R32-based device	Accuracy on synthetic dataset	Achieves target accuracy on synthetic data	Synthetic dataset coverage	Article XXXVII: SYNTHETIC vs REAL LOOP
WP-02	ML model trained on synthetic data	Computer	Cross-validation accuracy	Achieves target accuracy on synthetic data	Performance on synthetic data	Article XXXVIII: NO REAL DATA
WP-03	P-13 ML predictor: Real failure data (if available)	Real device	Real failure rate	Sufficient real failure events	Real failure data	Article XXXIX: CRITICAL UNKNOWN FOR REAL DATA
WP-04	Model retrained on real data (if available)	Computer	Accuracy on real hold-out data	Achieves target accuracy on real data	Real data training results	Article XL: REAL DATA IMPROVEMENT
WP-05	P-13 ML predictor: Software as a Medical Device (SaMD)	Software on R32-based device	EC 62304 V&V compliance	EC 62304 V&V compliance	EC 62304 V&V report	46 weeks
WP-06	Clinical validation study	Clinical sites	Behavioral sensitivity + specificity	Clinically useful accuracy	Clinical validation results	12-24 months (clinical timeline)

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 6-step build plan. The next experiment is clearly specified. Target buyer: Medical device AI/digital health company; neuroshunt manufacturer with digital platform.

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: No real failure dataset exists to train the model (CRITICAL BLOCKER per Article XXXVII)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: No real failure dataset exists to train the model (CRITICAL BLOCKER per Article XXXVII)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.